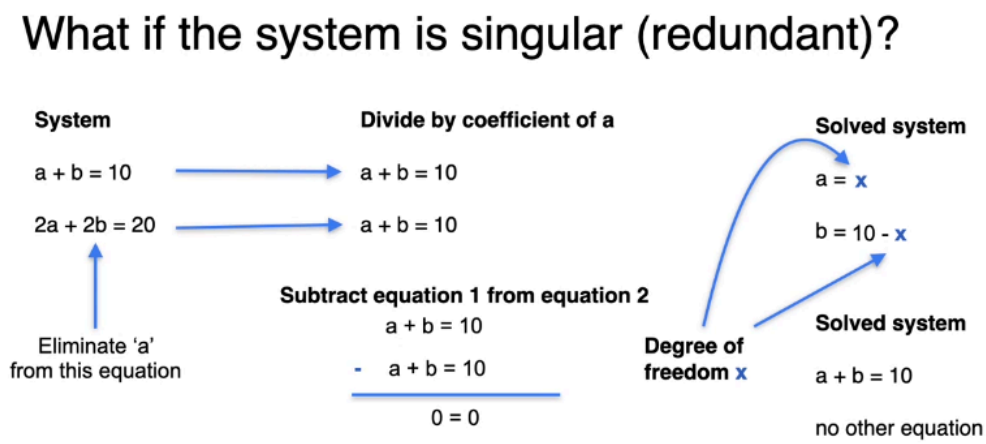
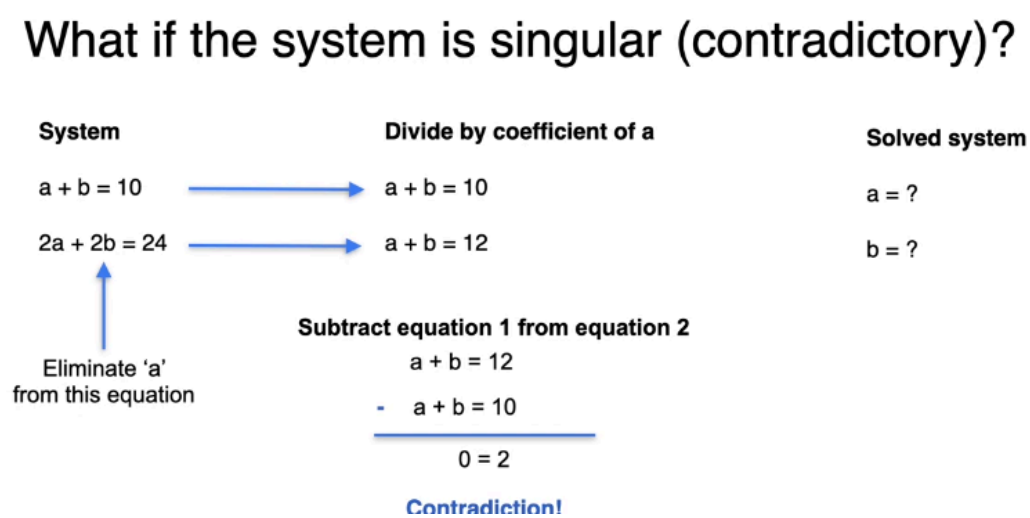
I. Solving non-singular system of linear equations

I already know how

II. Solving singular system of linear equations





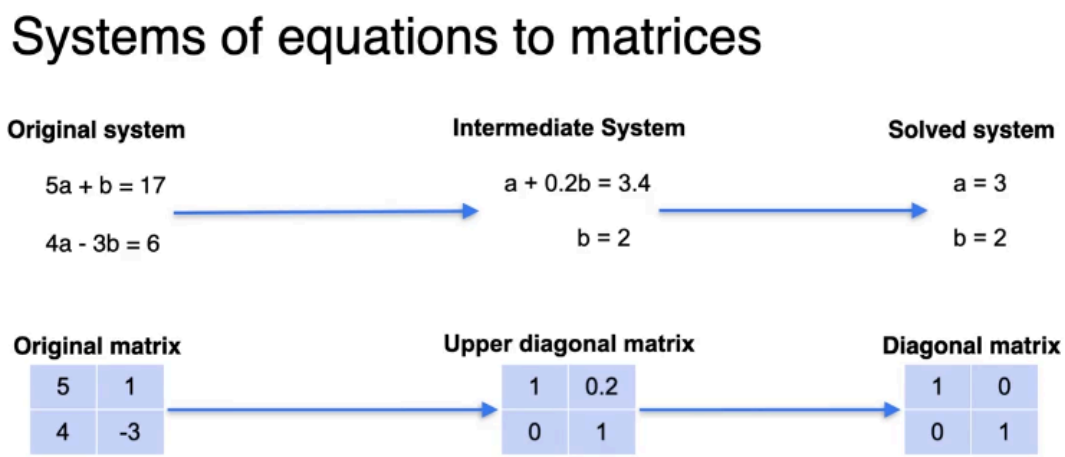
III. Solving system of equations with more variables

Already i know how

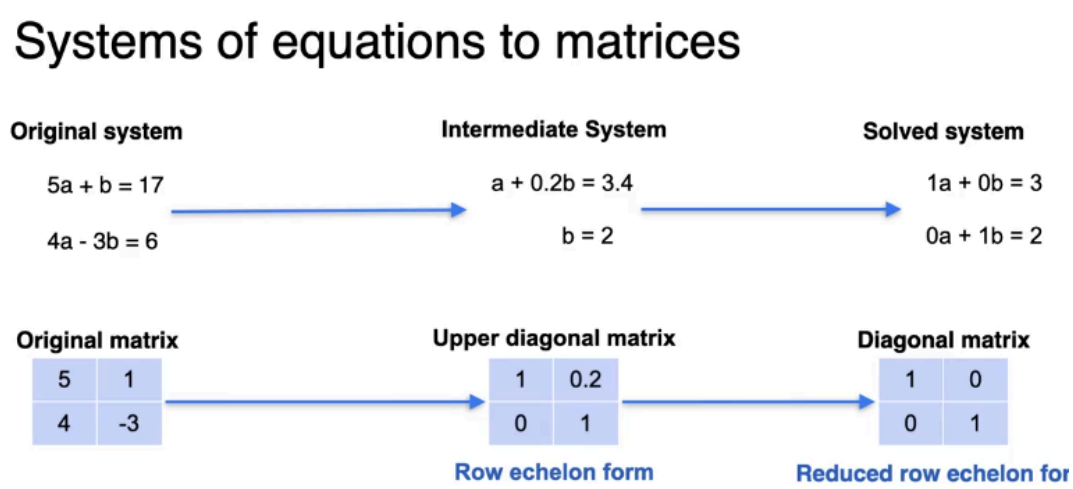
IV. Matrix row-reduction

If you know how to solve systems of linear equations by manipulating equations, then you already understand the idea behind **matrix row reduction**. Matrix row reduction—also called **Gaussian elimination**—uses the **same algebraic operations**, but applies them directly to the **rows of a matrix** instead of individual equations.

The goal is to transform the matrix into a simpler form from which solutions and other useful information can be easily extracted.



A system like a=3 and b=2 can be rewritten as 1a+0b=3 and 0a+1b=2



Form of row echelon form

A matrix in **row echelon form** has a very specific structure. Along the **main diagonal**, there are some **1s**, possibly followed by **zeros**. Everything **below the diagonal is zero**. To the **right of each leading 1**, any numbers are allowed, but once a diagonal entry is zero, **all entries to its right must also be zero**.

For a **2×2 matrix**, this means only three cases are possible:

- **Two 1s on the diagonal** → full rank
- **One 1 on the diagonal** → reduced rank
- **Zero 1s on the diagonal** → zero matrix

These cases help classify systems of equations and determine how much information the matrix carries.

V. Row operations that preserve singularity

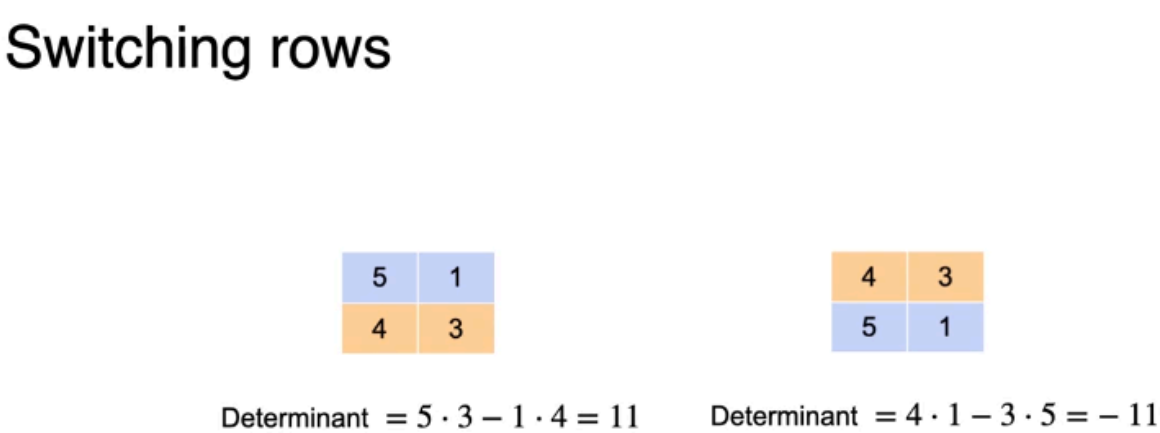
Before diving into row reduction, it's important to understand **row operations**. These are the same manipulations you use when solving systems of linear equations, now applied directly to **rows of a matrix**.

A key property of row operations is that they **preserve singularity**:

- Applying them to a **non-singular matrix** keeps it non-singular.
- Applying them to a **singular matrix** keeps it singular.

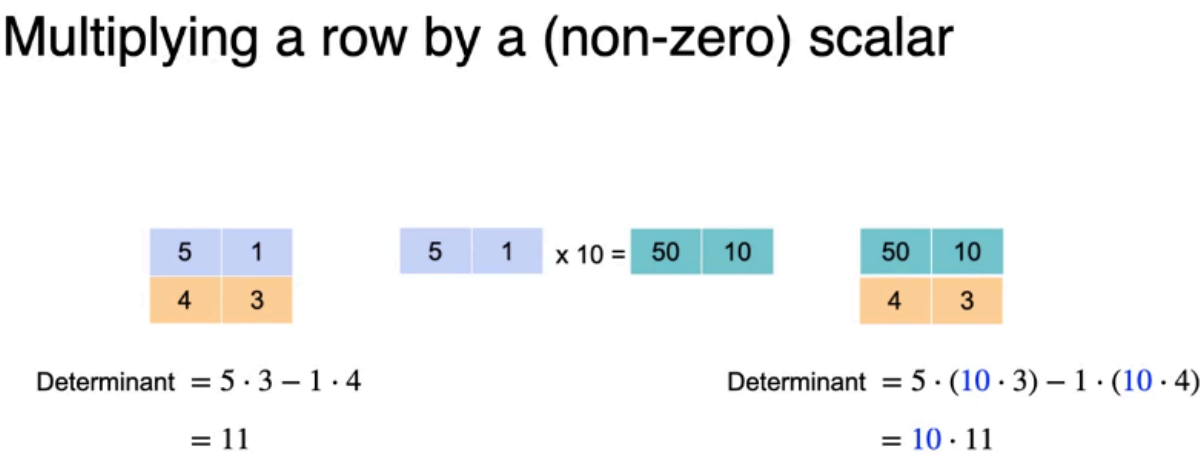
V.1. Row switching

row swapping flips the sign of the determinant but does not change whether the matrix is singular or non-singular.



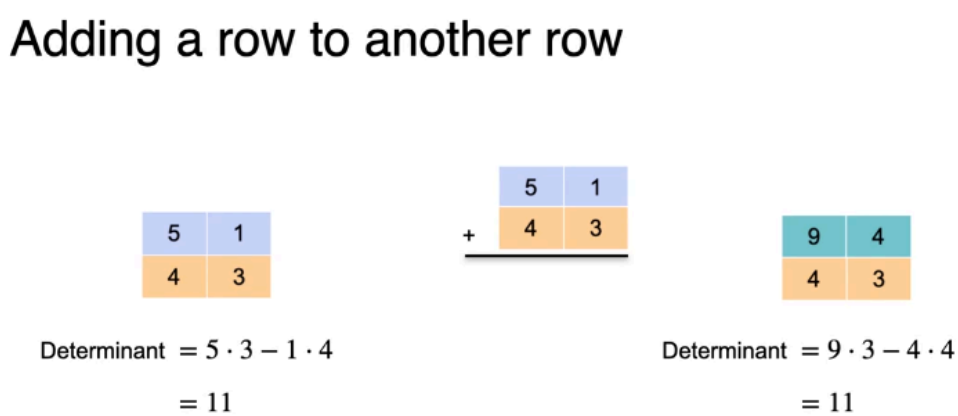
V.2. Multiplying a row by a (non-zero) scaler

Multiplying a row by a non-zero number keeps a non-zero determinant non-zero and a zero determinant zero. Therefore, this row operation **preserves singularity and non-singularity**.



V.3. Adding a row to another

Believe it or not his operation **preserves the determinant** and therefore also **preserves singularity and non-singularity**.



VI. The rank of a matrix

VI.1. Systems of information

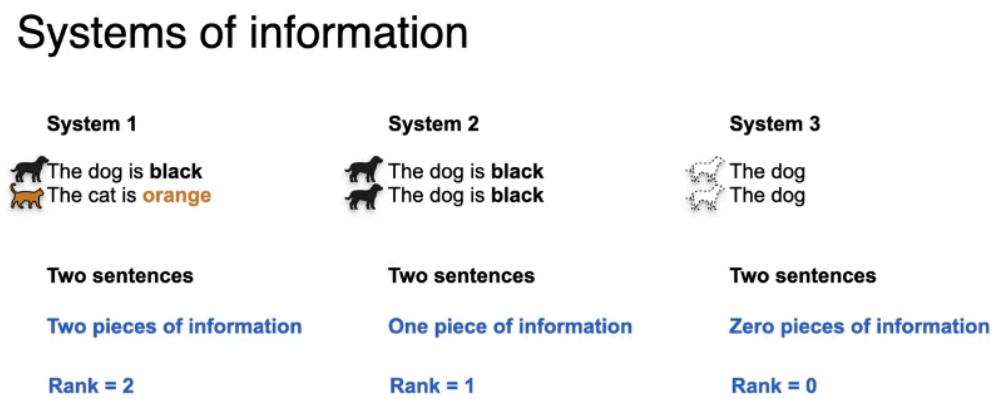
Throughout the previous lessons, you've been using an idea without naming it: **the rank of a matrix**. It is the notion of the rank of a matrix which in some way measures how much information that matrix or its corresponding system linear equations is carrying.

System 1: "The dog is black" and "The cat is orange."
This system has two different sentences and gives **two independent pieces of information**, so its **rank is 2**.

System 2: "The dog is black" and "The dog is black."
Although there are two sentences, they repeat the same fact. The system carries **only one piece of**

information, so its **rank is 1**.

System 3: Two contradictory or meaningless statements about the dog.
These sentences provide **no useful information** to determine the colors, so the **rank is 0**.



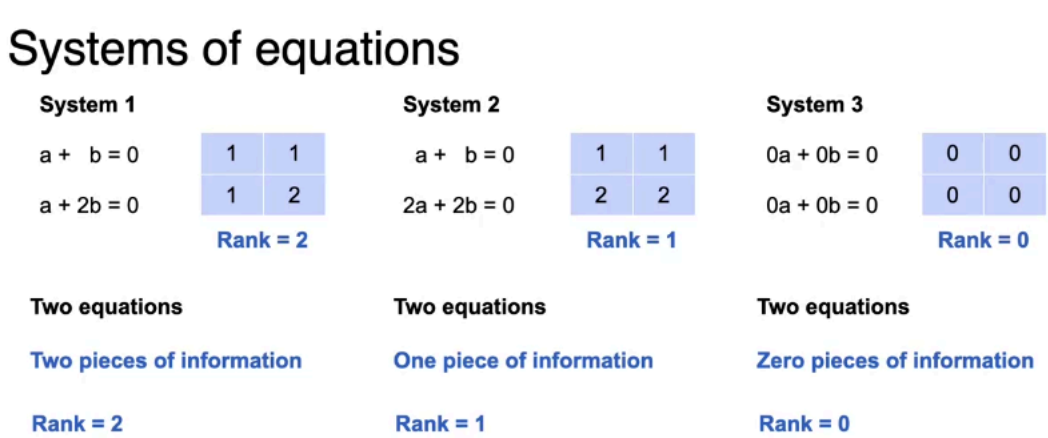
VI.2. Systems of equation

- In the **first system**, each equation adds new information. One equation restricts solutions to a line, and the second narrows that line to a single point. The system carries **two pieces of information**, so its **rank is 2**.
- In the **second system**, the two equations are actually the same. Only one equation contributes information, narrowing solutions to a line. The system carries **one piece of information**, so its **rank is 1**.
- In the **third system**, the equations provide no restrictions—any values satisfy them. The system carries **zero information**, so its **rank is 0**.

Every system of linear equations has a **corresponding matrix**, and the **rank of a matrix** is defined as the **rank of its corresponding system of equations**.

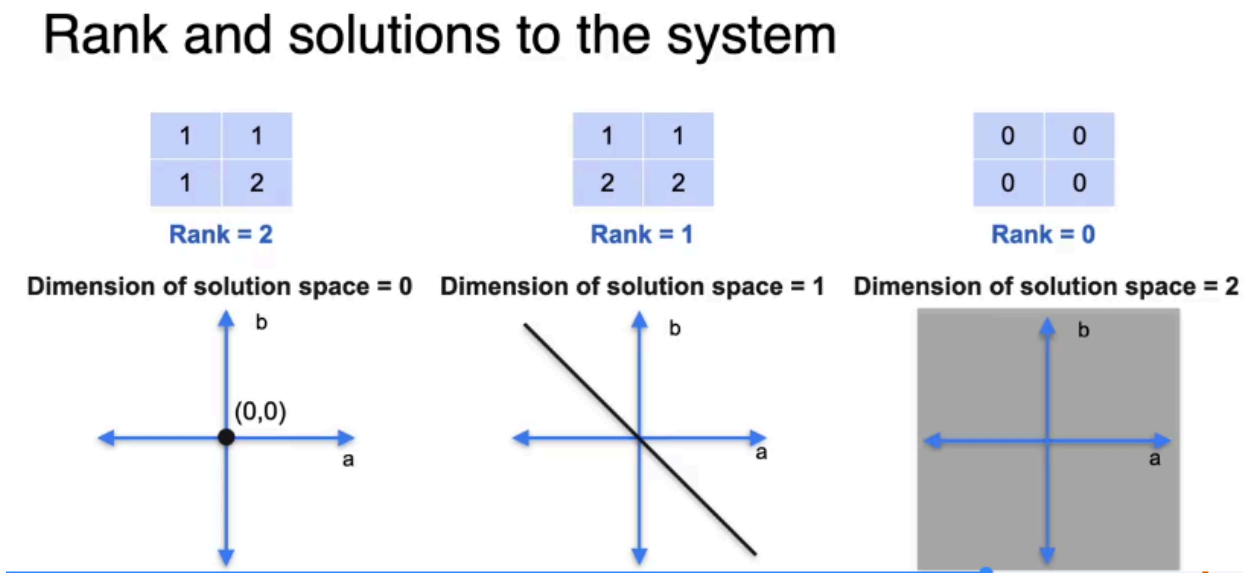
So:

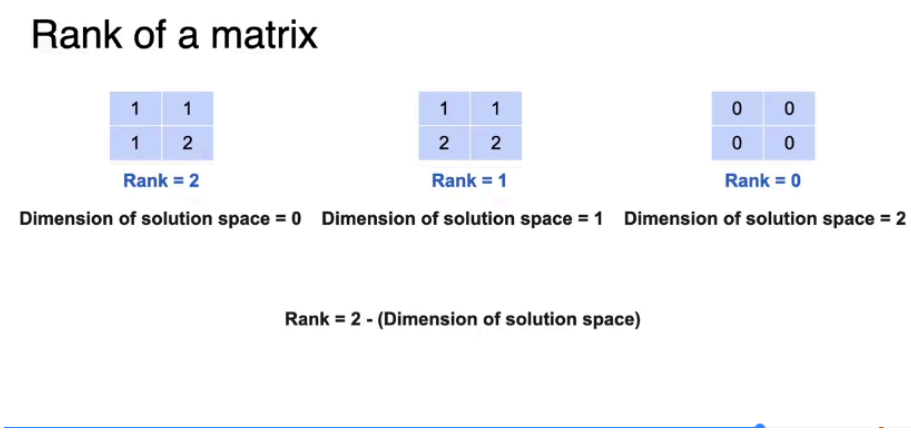
- Matrix of system 1 → **rank 2**
- Matrix of system 2 → **rank 1**
- Matrix of system 3 → **rank 0**



VI.3. Rank and solutions to the system

(x,y) = 0d line = 1D plane = 2D

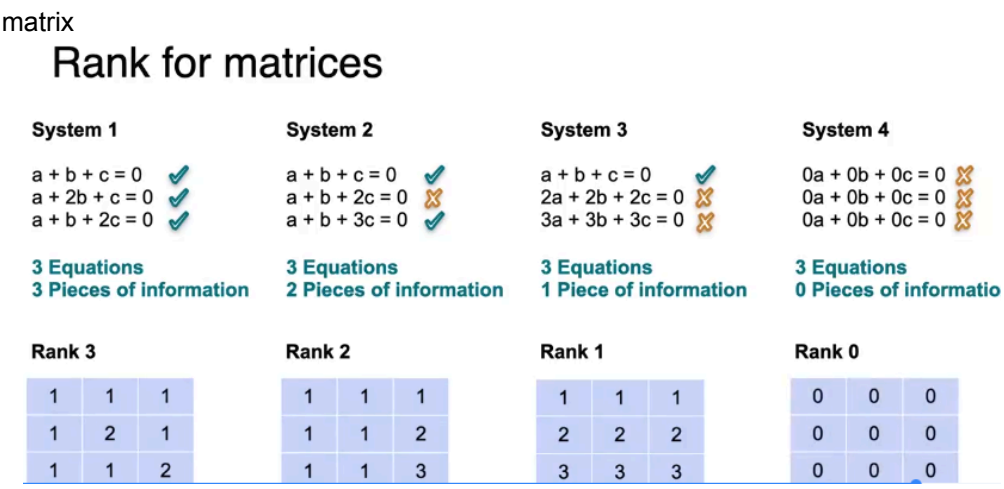




Note that for Non-singular matrix or system of sentences always Number of rows = rank because each row is a distinct information.

VII. The rank of a matrix in general

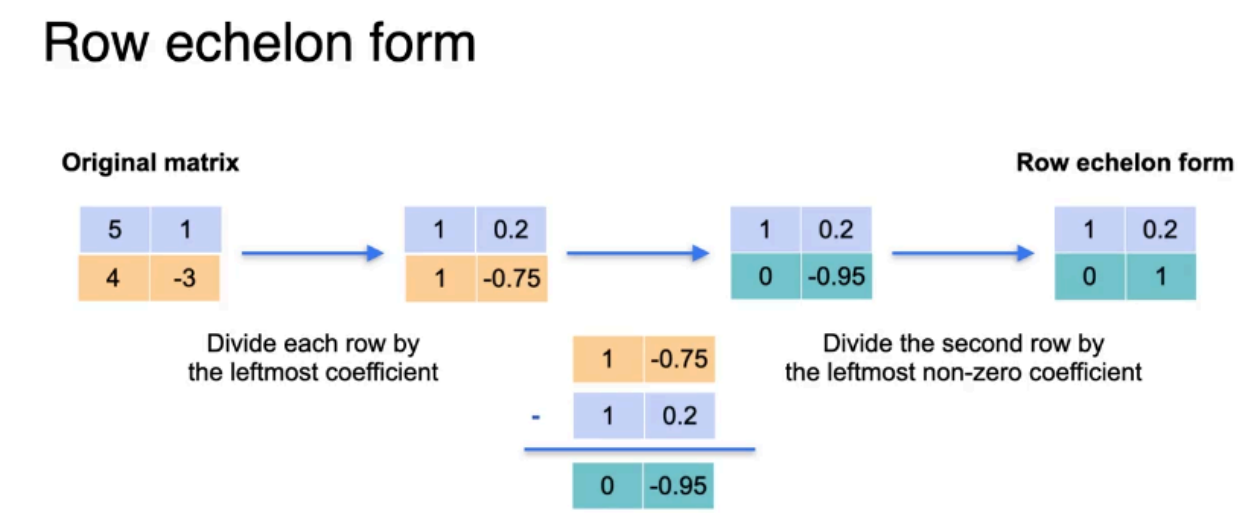
Rank of 3*3 matrix

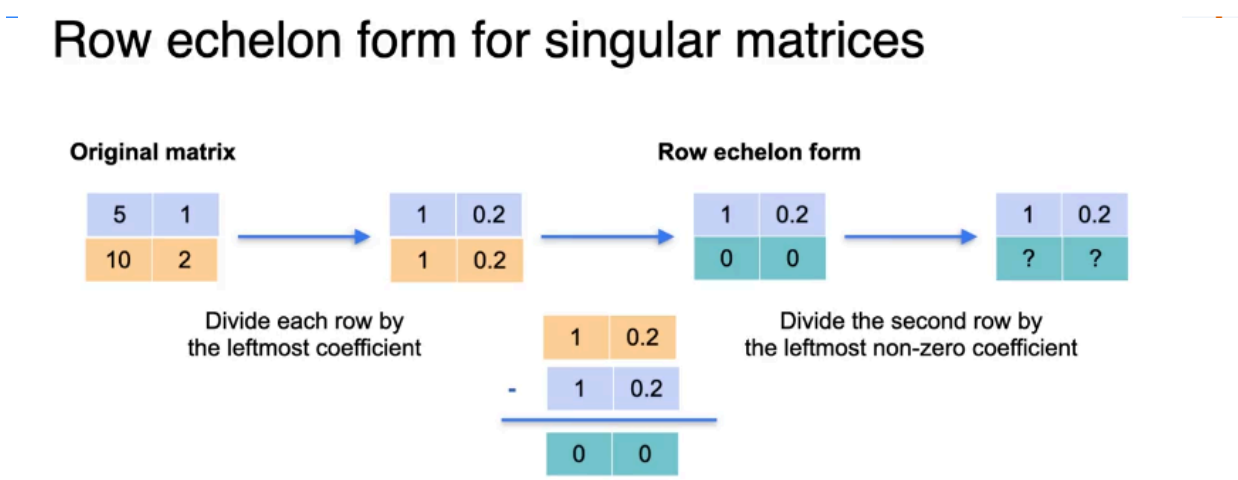


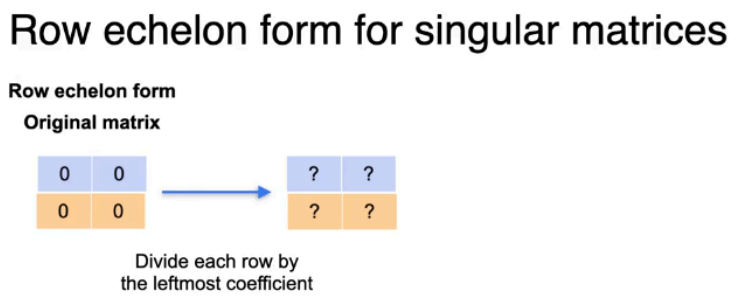
There is a question. It seems that calculating the rank is not that easy, so is there a simpler way to calculate the rank? The answer is yes, and it has to do with the row echelon form of a matrix.

VII. Row echelon form

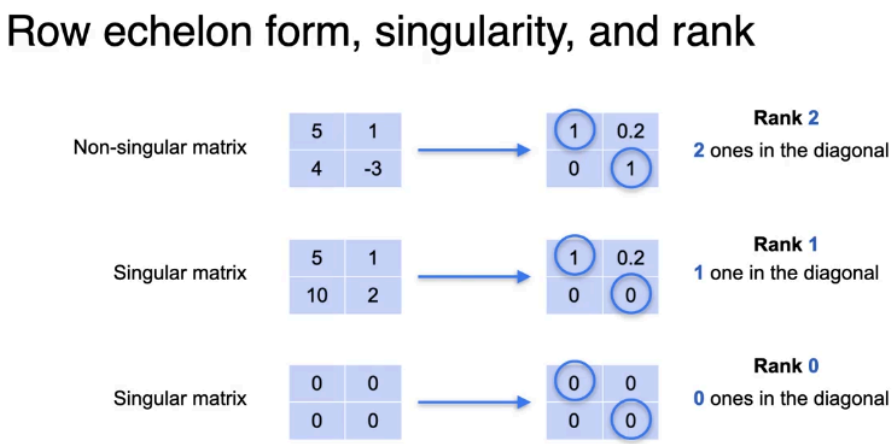
VII.1 How to get row echelon form from a matrix ?







VII.2. Row echelon form and singularity and rank



The rank of a matrix is the number of ones in the diagonal of the row echelon form.

"A matrix is NON-singular if and only if there are only ones and no zeros in the diagonal."

VIII. Row echelon form in general

General informations:

- Rows **may or may not be all zeros**, but any **zero rows must go at the bottom**.
- Every **non-zero row has a leftmost non-zero entry**, called a **pivot**.
- **Pivot rule:** Each pivot must be **strictly to the right** of the pivot in the row above. In other words, no pivot can be in the same column or to the left of the pivot above it.

The **row echelon form (REF)** is very useful because the **rank of the matrix** equals the **number of pivots**.

In **row echelon form (REF)**, all entries **below the pivots** (which often lie along a “stair-step diagonal”) must be **zeros**.

- The pivots don’t have to be 1 in REF—they can be any non-zero number.
- The key is that in each column that contains a pivot, **all entries below that pivot must be zero**.

Row echelon form

2	*	*	*	*
0	1	*	*	*
0	0	3	*	*
0	0	0	-5	*
0	0	0	0	1

Rank 5

3	*	*	*	*
0	0	1	*	*
0	0	0	-4	*
0	0	0	0	0
0	0	0	0	0

Rank 3

- Zero rows at the bottom
- Each row has a pivot (leftmost non-zero entry)
- Every pivot is to the right of the pivots on the rows above
- Rank of the matrix is the number of pivots

Important note: notation about what he will use in this course

Important note

3	*	*	*	*
0	0	1	*	*
0	0	0	-4	*
0	0	0	0	0
0	0	0	0	0

$\xrightarrow{\div 3}$ $\xrightarrow{\div 1}$ $\xrightarrow{\div(-4)}$

1	*	*	*	*
0	0	1	*	*
0	0	0	1	*
0	0	0	0	0
0	0	0	0	0

Note:
- In general, pivots different than 1 are allowed
- For this class, pivots are 1. This makes no mathematical difference.

Examples:

Rank for matrices

Matrix 1

1	1	1
1	2	1
1	1	2

Matrix 2

1	1	1
1	1	2
1	1	3

Matrix 3

1	1	1
2	2	2
3	3	3

Matrix 4

0	0	0
0	0	0
0	0	0

Row echelon forms

1	1	1
0	1	0
0	0	1

Number of pivots = 3

1	1	1
0	0	1
0	0	0

Number of pivots = 2

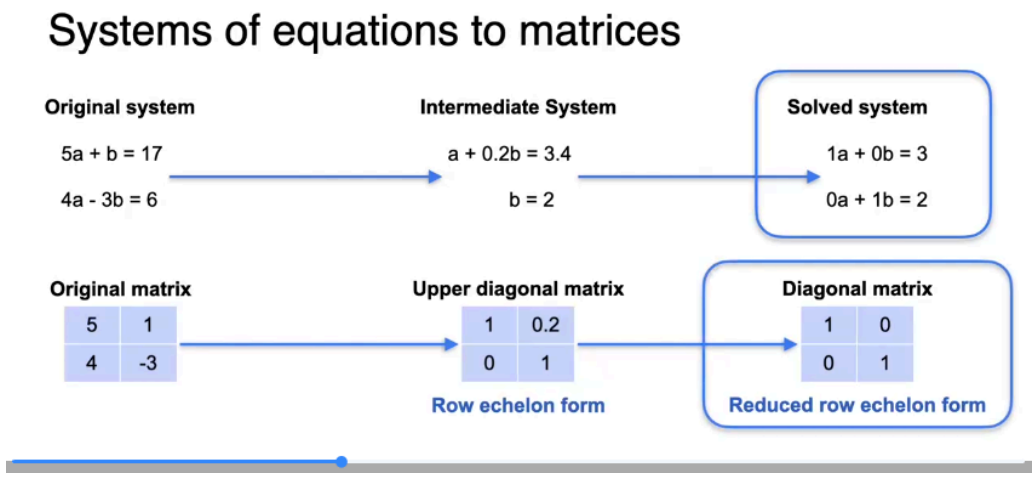
1	1	1
0	0	0
0	0	0

Number of pivots = 1

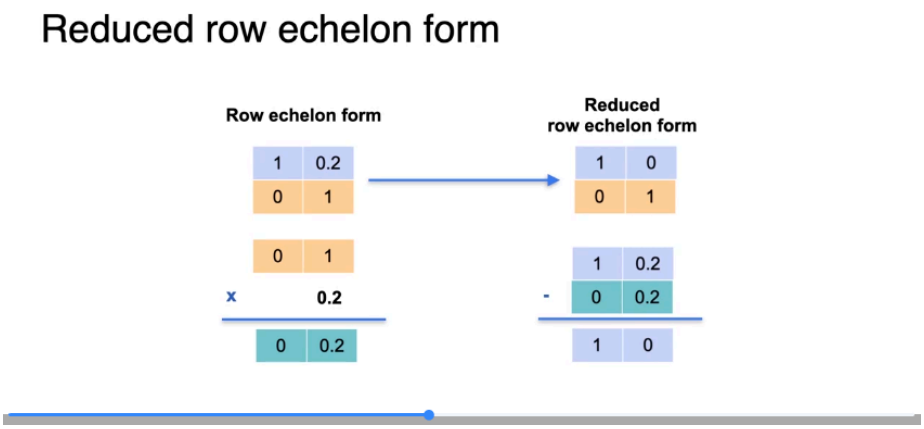
0	0	0
0	0	0
0	0	0

Number of pivots = 0

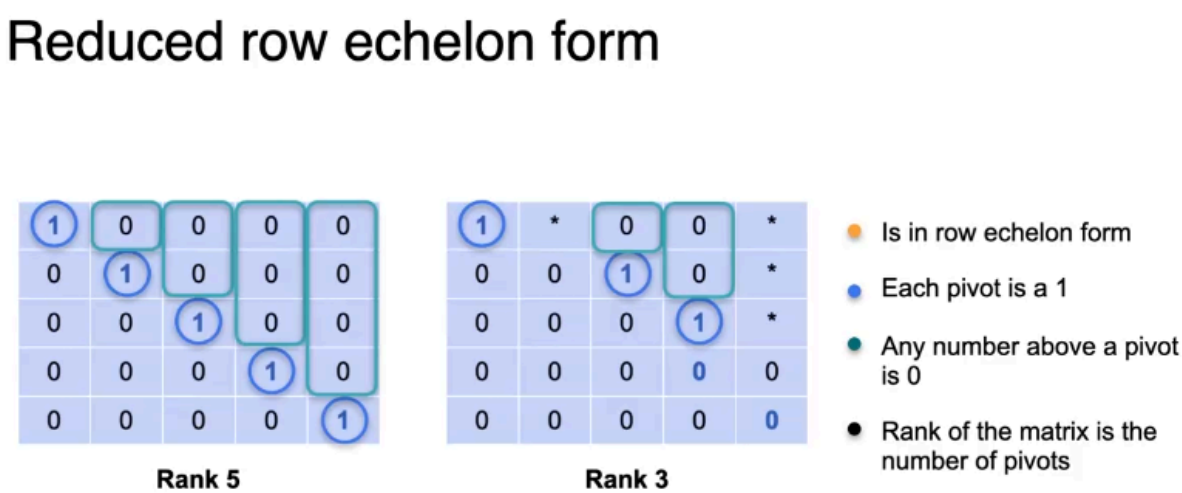
IX. Reduced row echelon form



How to get reduced row echelon form ?



Looks of reduced row echelon form

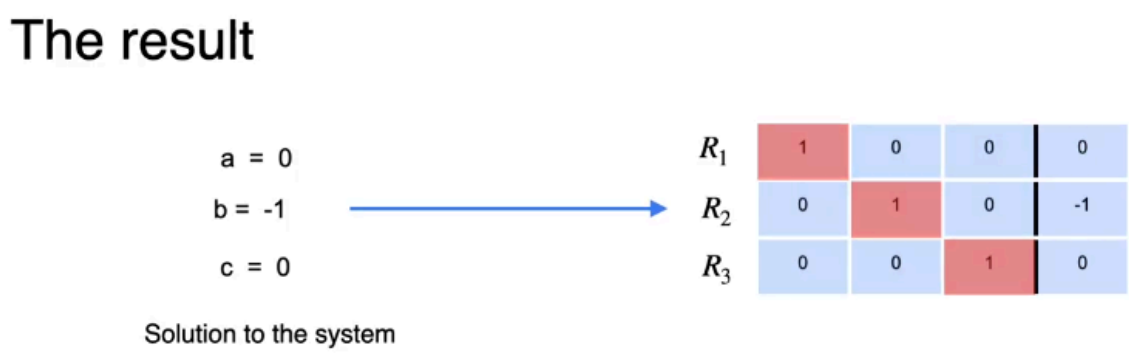


X. The Gaussian Elimination Algorithm

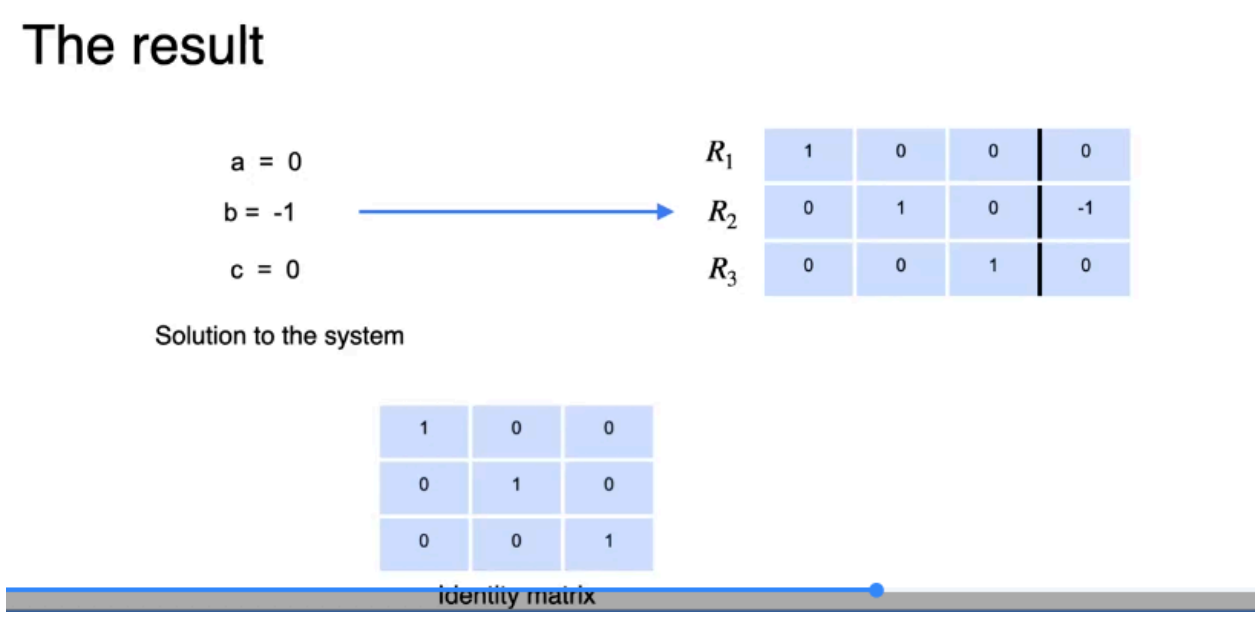
You can find the algorithm using direct method of sympy library or using numpy.

Gaussian algorithm :

System -> Matrix -> augmented matrix -> pivoting -> back-substituion -> result(solution to the system)



When we reach to the identity matrix means we have solve the equation



What if the system is singular ? When we get a whole rows of 0 that means we need to stop and the matrix is singular

Checking if it has infinitely many or no solutions

- Row full of zeroes in row echelon form
- Constant in that row is zero
- **Infinitely many solutions**

1	2	-1	5
0	0	-7	9
0	0	0	0

- Row full of zeroes in row echelon form
- Constant in that row is not zero
- **No solutions**

1	2	-1	5
0	0	-7	9
0	0	0	4

SUmmary ;

Gaussian Elimination - Summary

1. Create the augmented matrix
2. Get the matrix into reduced row echelon form
3. Complete back substitution
4. Stop if you encounter a row of 0s